

Cybersecurity Task-2 Cybersecurity & Ethical Hacking Lab Setup Using Kali Linux and DVWA

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1. ABSTRACT

This project focuses on setting up a cybersecurity laboratory using Kali Linux and Damn Vulnerable Web Application (DVWA) in a virtual environment. The lab provides a safe platform for learning ethical hacking, network analysis, and web application security testing. Tools such as Nmap, Burp Suite, and Wireshark were used to perform network scanning, traffic interception, and packet analysis. The project helped in understanding network communication, vulnerability assessment, and the practical use of cybersecurity tools. The developed lab serves as a foundation for advanced penetration testing and security research. The successful implementation of this laboratory provided practical exposure to network reconnaissance, web application interaction, packet inspection, and security assessment methodologies.

KEYWORDS: - Cybersecurity, Ethical Hacking, DVWA, Nmap, Wireshark, Network Security, Burp Suite, Web Application Security

2. INTRODUCTION

Cybersecurity plays a vital role in protecting computer systems, networks, and sensitive information from cyber threats. Practical knowledge is essential for understanding security concepts and attack methodologies. This project involves creating a virtual cybersecurity lab using Kali Linux as the attacker machine and DVWA as the target machine. The environment allows safe testing of security tools such as Nmap, Burp Suite, and Wireshark. Through this project, users gain hands-on experience in network scanning, traffic monitoring, and web application security assessment while working in an isolated and controlled environment.

3. OBJECTIVE

- The objective of this project is to build a safe and isolated cybersecurity laboratory environment for learning and practicing ethical hacking techniques.
- The lab consists of a attacker machine running Kali Linux and a vulnerable web application (DVWA) deployed as the target system.
- This environment enables the safe execution of security assessments, web application testing, network analysis, and traffic monitoring without affecting real-world systems.

4. TOOLS AND TECHNOLOGIES USED

- ✓ Kali Linux
- ✓ DVWA (Damn Vulnerable Web Application)
- ✓ VirtualBox
- ✓ Nmap
- ✓ Burp Suite
- ✓ Wireshark

5. LAB ARCHITECTURE

- The Cybersecurity lab consists of two systems:
 - (1) Attacker Machine – Kali Linux
 - (2) Target Machine – DVWA Web Application
- Network Configuration:
 - NAT Adapter: Provides internet access for updates and package installation.
 - Host-Only Adapter: Enables communication between Kali Linux and DVWA while keeping the lab isolated from external networks.
- This setup ensures a secure environment for performing vulnerability assessments and traffic analysis.

6. INSTALLATION AND CONFIGURATION

6.1 Kali Linux Setup

- Imported Kali Linux virtual machine.
- Allocated 4 GB RAM and 2 CPU cores.
- Configured two network adapters: NAT and Host-Only
- Updated the operating system using apt update and upgrade.



Figure 1. Kali Linux Virtual Setup

6.2 DVWA Setup

- Deployed DVWA inside the web server directory.
- Verified successful access through a web browser.

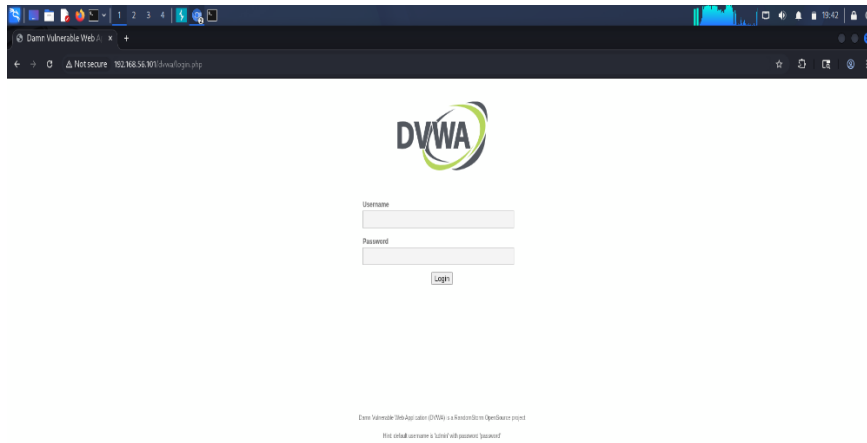


Figure 2. DVWA Login Interface

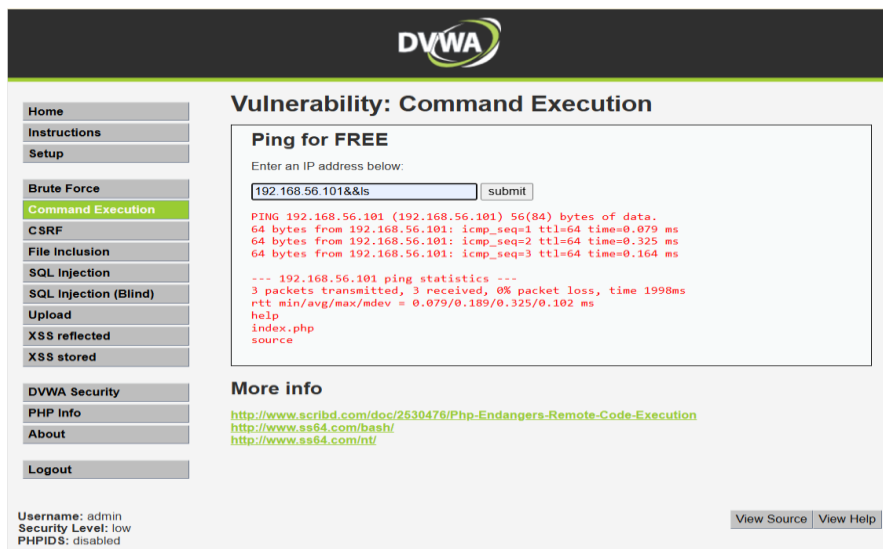
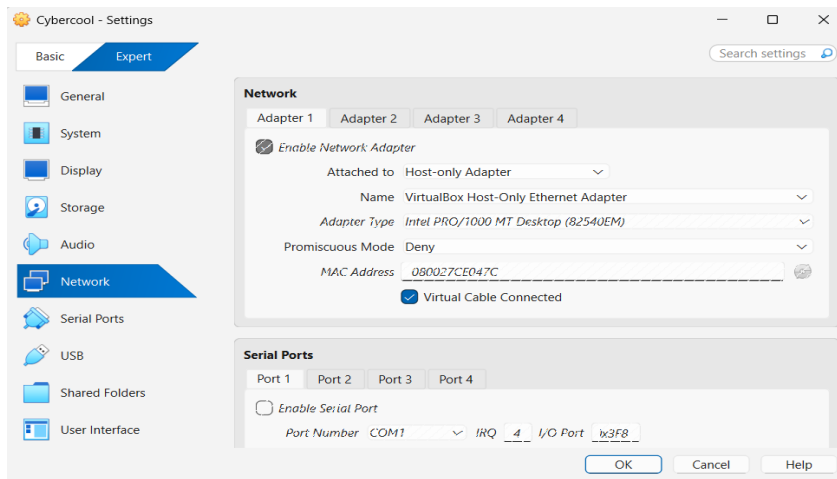


Figure 3. Command Execution in DVWA

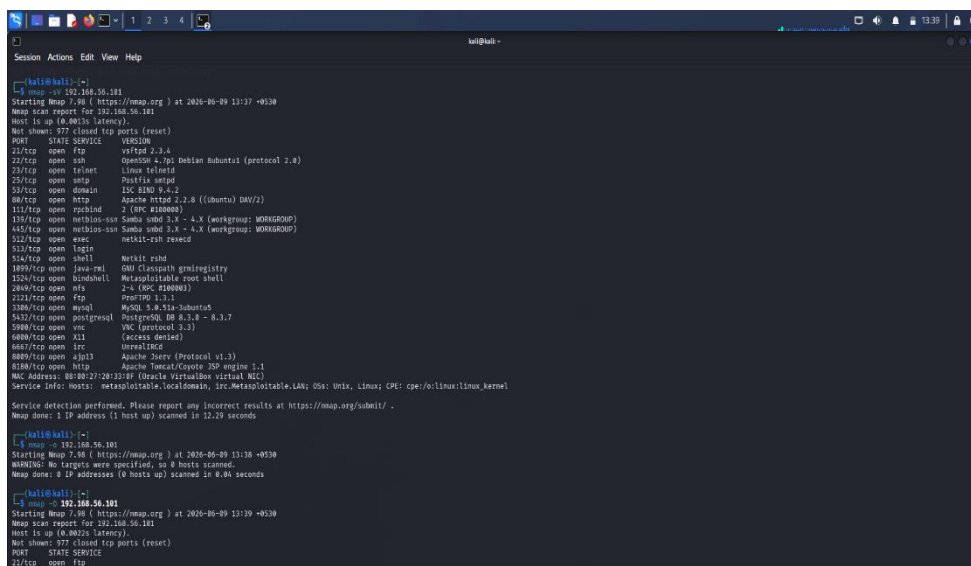
7. NETWORK VALIDATION

- ❖ After configuring the lab environment, network connectivity between the attacker and target systems was verified.
- ❖ The ping utility was used to confirm communication between Kali Linux and the DVWA host.
- ❖ Result:
 - Successful ICMP replies received.
 - Network connectivity established.



8. Nmap SCANNING

- ✓ Nmap was used to discover open ports and identify running services on the target machine.
- ✓ Command used: `nmap -sV`
- ✓ Findings:
 - Open HTTP service detected on Port 80.
 - Service versions identified successfully.
 - Target host discovered and accessible.
- ✓ Purpose:
 - Host discovery
 - Port scanning
 - Service enumeration



```

kali@kali:~$ nmap -sS 192.168.56.101
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-09 13:29 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0025s latency).
Not shown: 277 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ftp
27/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
112/tcp   open  rshind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  nmap
513/tcp   open  login
514/tcp   open  shell
3890/tcp  open  mmsregistry
3526/tcp  open  ingreslock
2800/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5908/tcp  open  vnc
6000/tcp  open  x11
6067/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  tomcat
Nmap Address: 06:80:27:28:13:0F (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.1
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.98 seconds

```

Figure 6. Nmap Port Scanning Result 2

9. BURPSUITE ANALYSIS

- ❖ Burp suite was configured as an intercepting proxy between the browser and DVWA.
- ❖ Activities performed:
 - ➔ Intercepted HTTP requests.
 - ➔ Analyzed request and response headers.
 - ➔ Observed web application communication.
 - ➔ Verified proxy functionality.
- ❖ Result: Successful interception and forwarding of web traffic.

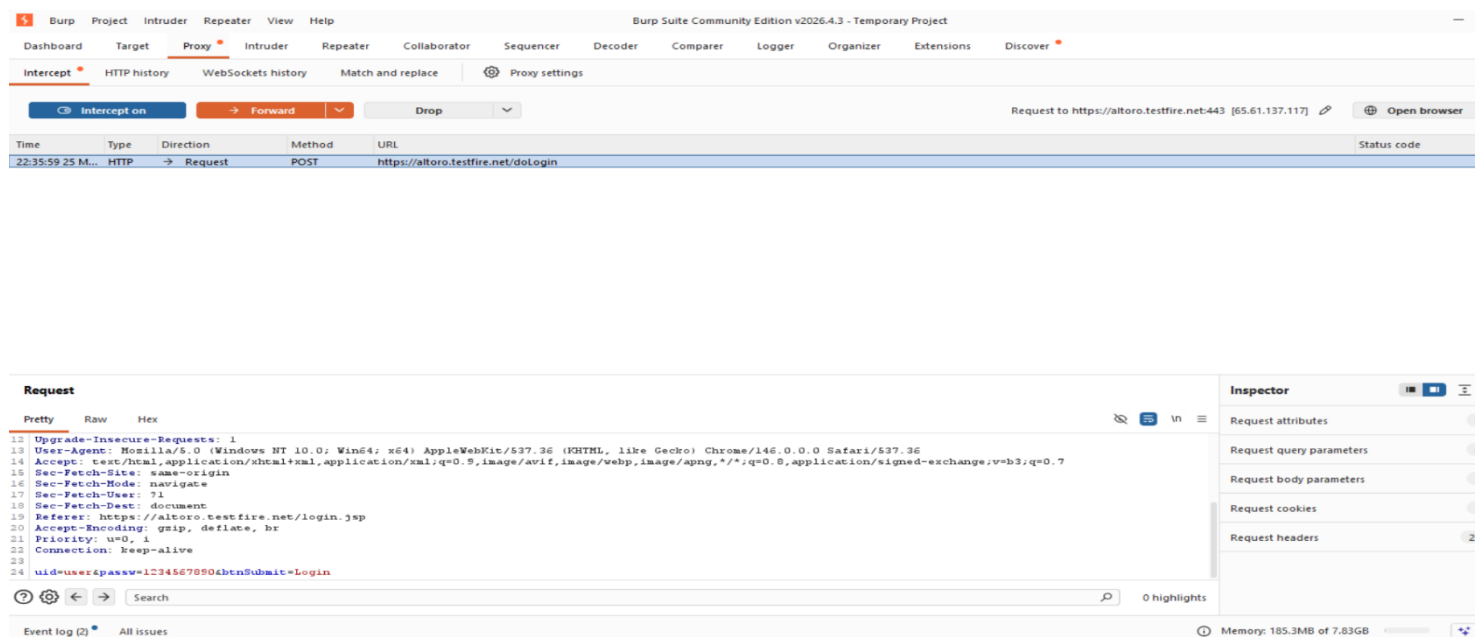


Figure 7. Burp Suite HHTTP Request Interception

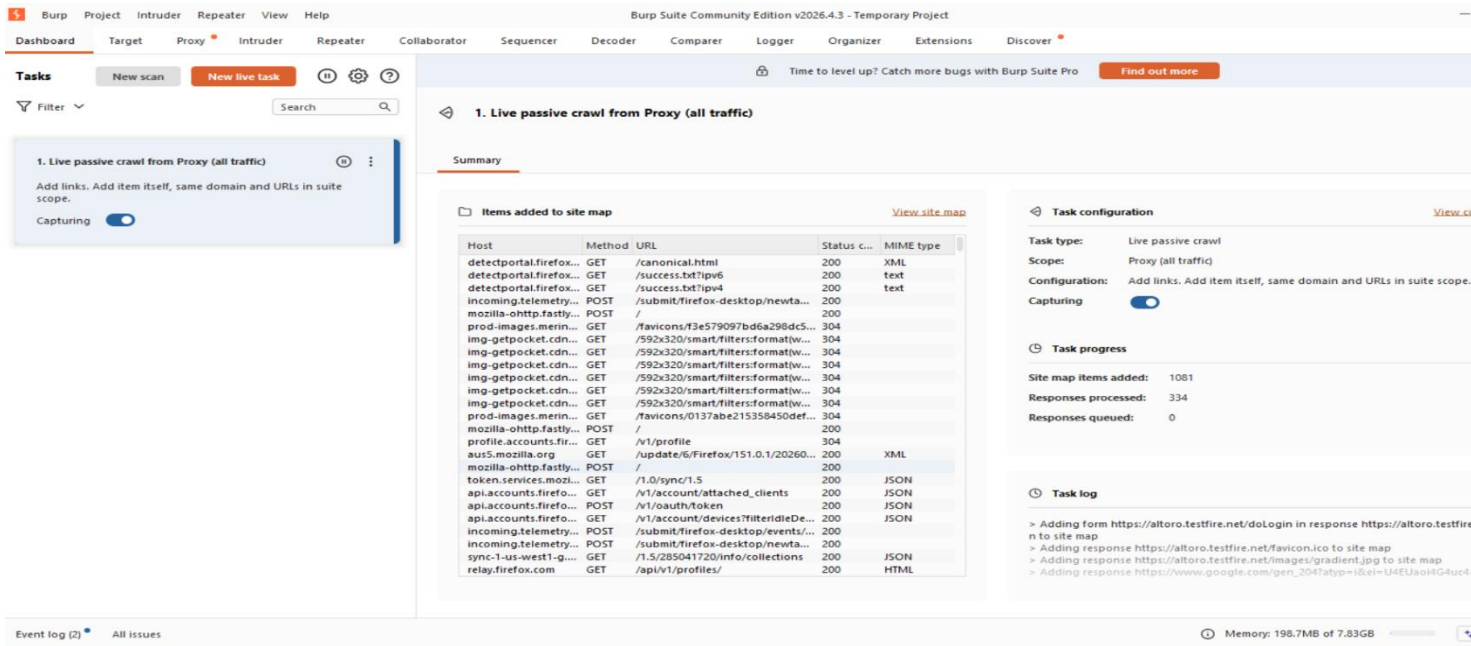


Figure 8. Burp Suite Dashboard

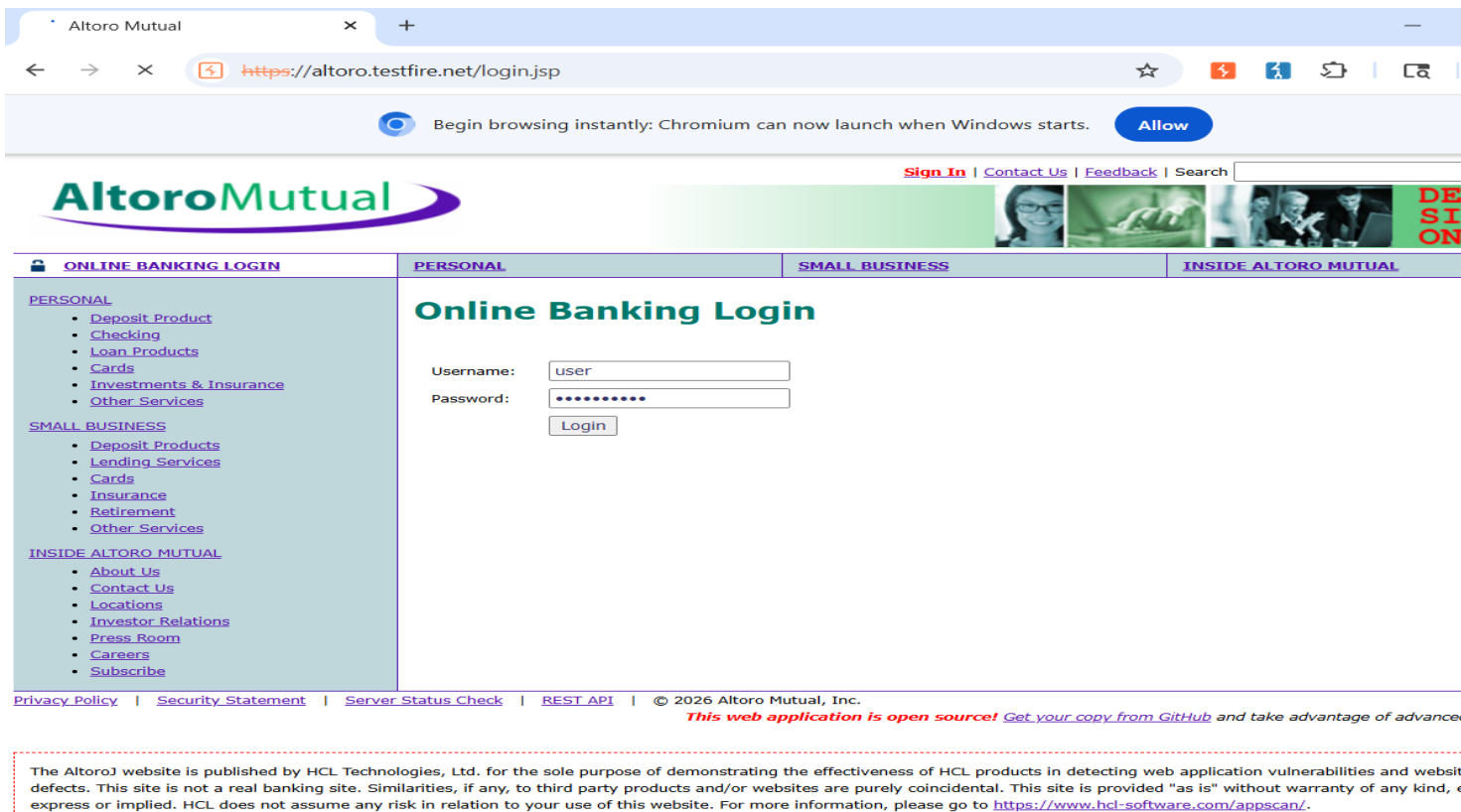


Figure 9. Burp Suite website login

10. WIRESHARK TRAFFIC CAPTURE

- Wireshark was used to monitor and analyze network traffic generated between Kali Linux and DVWA application.
- Activities performed:

- Filtered traffic using TCP and HTTP protocols.
- Examined packet details such as source IP, destination IP, and protocol information.

➤ Result:

- Successful capture of web traffic.
- Network communication visualized and analysed.

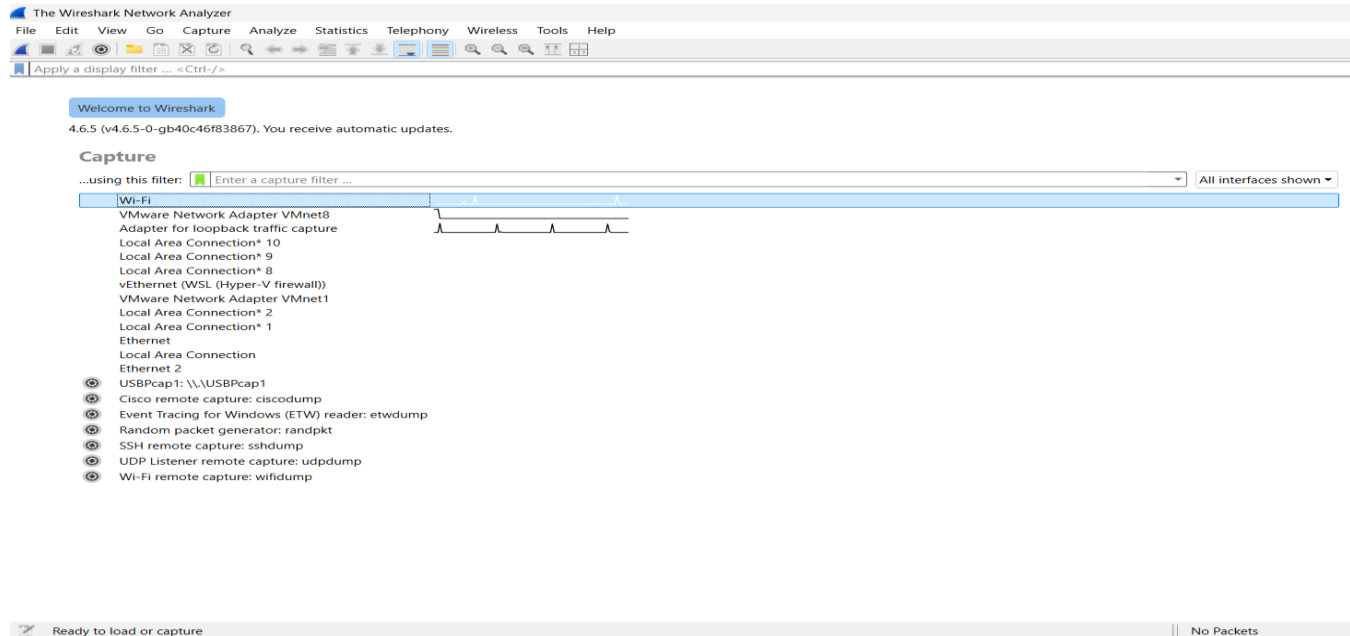


Figure 9. Wireshark interface

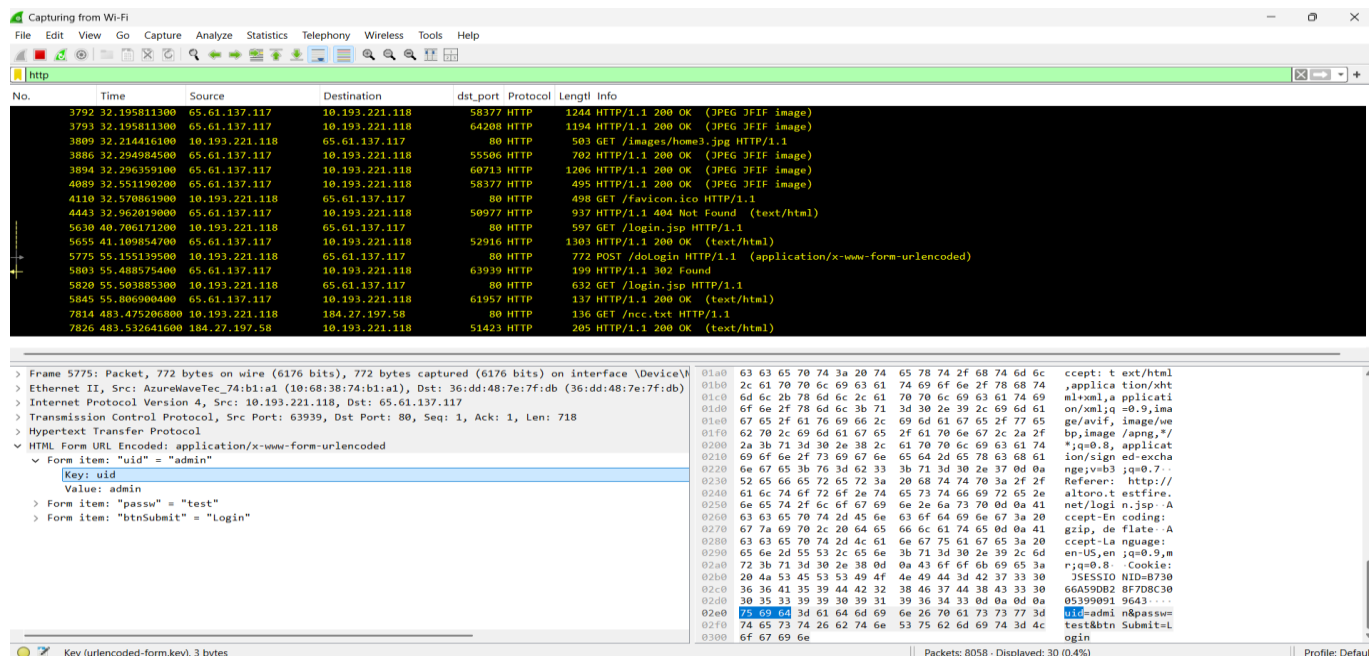


Figure 10. Wireshark Packet Capture Analysis

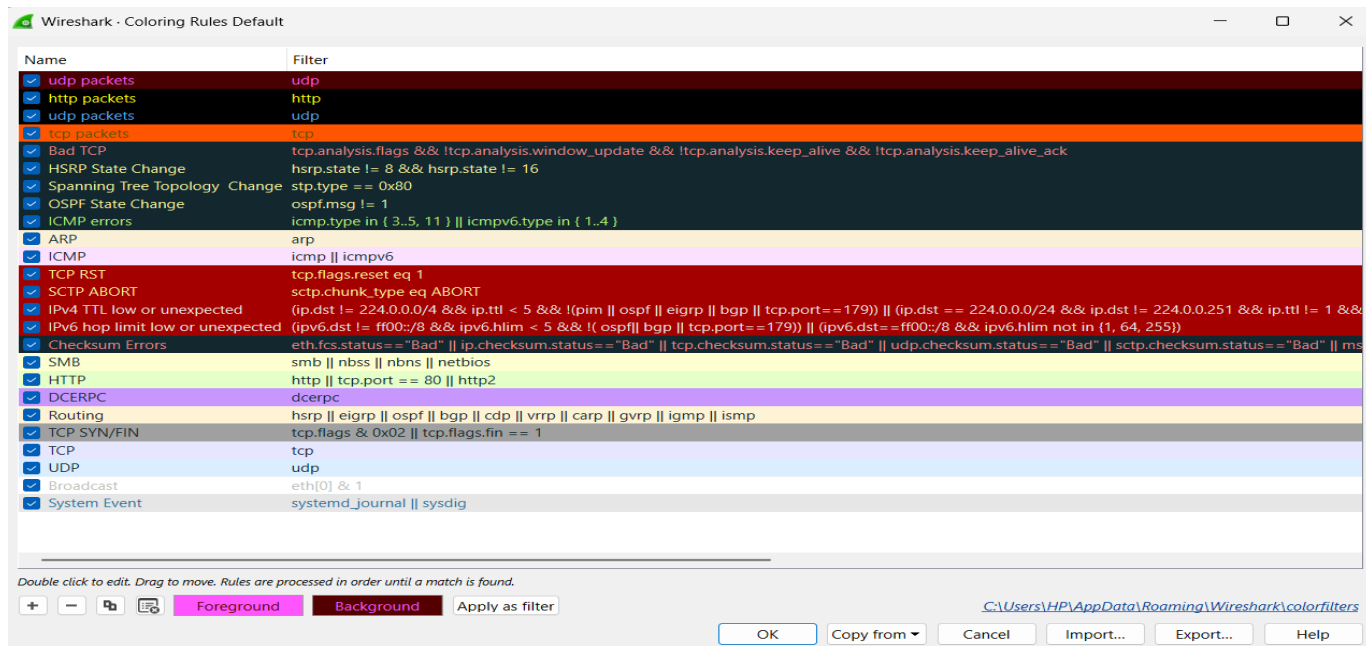


Figure 11. Wireshark Packet Capture Analysis

11. LEARNING OUTCOME

Through this project, the following skills were developed:

- a) Virtual machine deployment and management.
- b) Network configuration using NAT and Host-Only adapters.
- c) Basic network troubleshooting.
- d) Port scanning and enumeration using Nmap.
- e) Web traffic interception using Burp Suite.
- f) Packet analysis using Wireshark.
- g) Secure cybersecurity lab setup practices.

12. CHALLENGES FACED:

- 1) Configuring Host-Only networking.
- 2) Establishing communication between attacker and target machines.
- 3) Selecting the correct network interface in Wireshark.
- 4) Accessing DVWA from the attacker machine.
- 5) Capturing and filtering relevant network traffic.

13. CONCLUSION

The project successfully established a secure and isolated cybersecurity laboratory environment using Kali Linux and DVWA.

The environment allowed practical experience with network scanning, packet analysis, web application interaction, and security testing tools.

The lab serves as a foundation for advanced cybersecurity activities such as vulnerability assessment, web penetration testing, exploitation practice and incident analysis.